

Formal Verification finds and fixes an MCTP control-path bug in NVIDIA OpenSMA

ESBMC counterexample, confirmed and fixed upstream

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Full report: `verification/REPORT.md` | Issue: [NVIDIA/OpenSMA#1](#)

The bug — a four-line view

```
// pdk-mctp-platforms-router-plat.cpp:34
void set_cur_eid(RoutingTable& rt,
                 Packet::InterfaceType iface,
                 uint8_t eid)
{
    rt.ec.cur_eid.at(iface) = eid;  // (*)
}
```

Source: OpenSMA, © NVIDIA Corporation, Apache-2.0.

The asymmetric guard

- `cur_eid` is sized `UsEnd = 2`.
- Validator rejects only `iface ≥ End = 18`.
- Any `iface ∈ [2, 17]` **passes validation**, then aborts at (*).
- Sibling `get_cur_eid()` has the guard; `set_cur_eid()` **did not**.

Behaviour under production flags

With `-fno-exceptions -fno-rtti`, `std::array::at(00B)` calls `abort()` — the firmware terminates (SIGABRT, exit 134).

Trigger

An MCTP Control SetEpld Request whose `packet_interface` falls in the gap drives the production dispatch path into the OOB access.

What ESBMC proved — in three states

Setup

- Harness builds a Control SetEpld Request; 9 header fields constrained to pass `Validator::validate()`.
- `iface_val` left *nondeterministic* in `[2, 18)`.
- Calls production `on_set_endpoint_id()` unmodified.

Solver result

- 275 verification conditions; Bitwuzla returns SAT in **< 0.01 s**.

Production confirmation

- ESBMC-generated test case compiled and run natively $\Rightarrow$ **SIGABRT (signal 6)**.

ESBMC counterexample (abridged):

```
State 5    iface_val = 2
           (passes assume >= UsEnd && < End)

State 9    valid = 1
           (Validator::validate() returns true)

State 10   Violated property:
           std::array::at out of range
           [i::0 < 2 fails]
           at set_cur_eid()
           -> cur_eid.at(2) on size-2 array
```

VERIFICATION FAILED

End-to-end proof from validator entry through dispatch to OOB write — **no code modification, no manual test input**.

Outcome — confirmed and fixed upstream

Disclosure timeline

- Reported to NVIDIA as **NVIDIA/OpenSMA#1** with the ESBMC counterexample and a one-line fix.
- NVIDIA's OpenSMA team acknowledged the report and tracked the fix internally.
- **Fix applied upstream** (2026-05-11): the interface $\geq$ UsEnd bound check is now mirrored into `set_cur_eid()`, closing the asymmetric-guard gap.

Scope and caveats

The verification covered 22 selected modules of OpenSMA (~7,900 LOC of first-party C++); not the full codebase, and not vendored components (libspdm, FreeRTOS, MCU SDK glue). Findings beyond F-1 are documented in the full report; some are latent (no current packet-driven path) and recorded as such. **No claims of broad firmware insecurity are made or implied.**

Full report and reproducers: github.com/lucasccordeiro/NVIDIA-OpenSMA

ESBMC: github.com/esbmc/esbmc

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